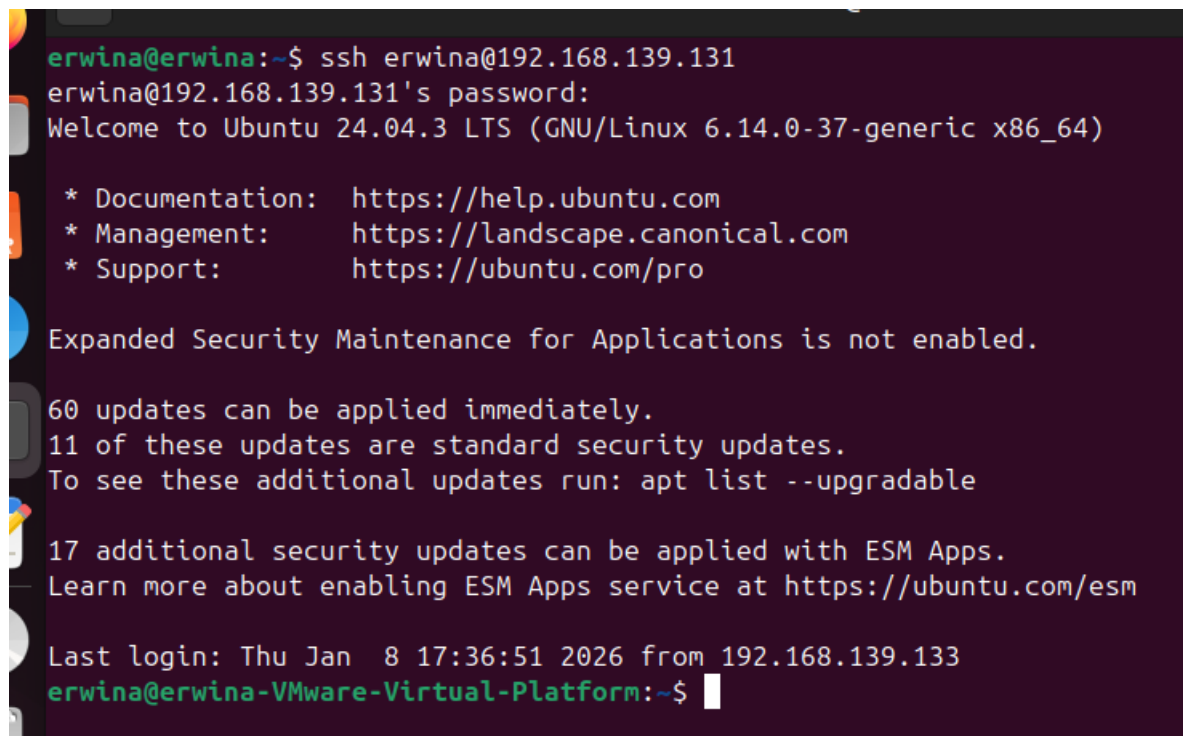


Template Week 6 – Networking

Student number: 583259

Assignment 6.1: Working from home

Screenshot installation openssh-server:



```
erwina@erwina:~$ ssh erwina@192.168.139.131
erwina@192.168.139.131's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

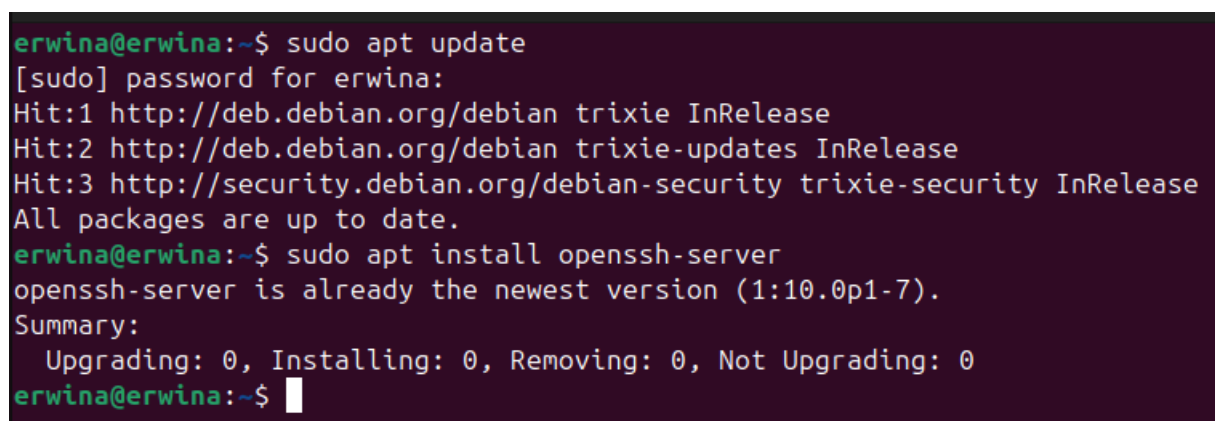
Expanded Security Maintenance for Applications is not enabled.

60 updates can be applied immediately.
11 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

17 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Thu Jan  8 17:36:51 2026 from 192.168.139.133
erwina@erwina-VMware-Virtual-Platform:~$
```

Screenshot successful SSH command execution:

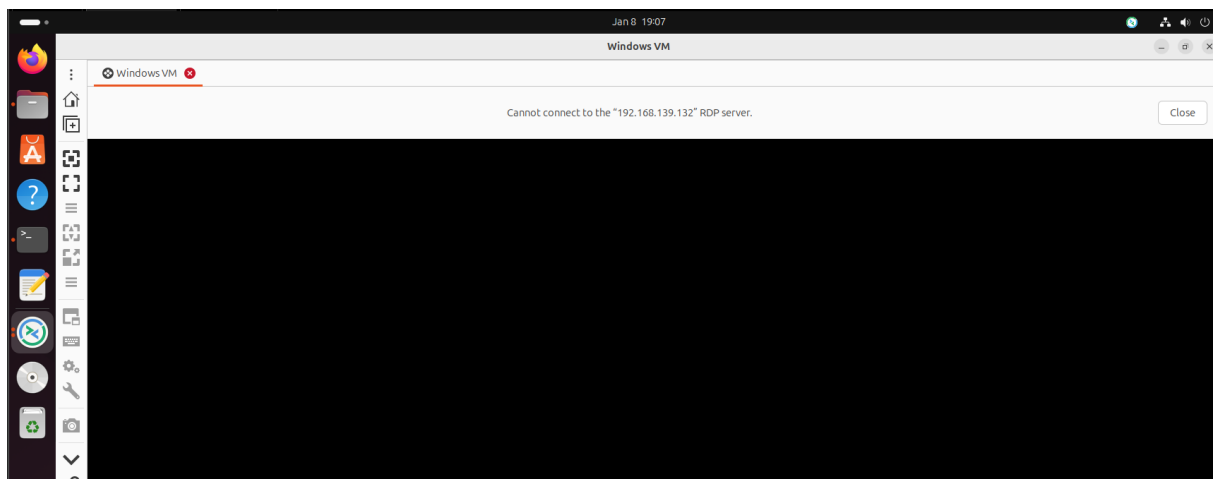


```
erwina@erwina:~$ sudo apt update
[sudo] password for erwina:
Hit:1 http://deb.debian.org/debian trixie InRelease
Hit:2 http://deb.debian.org/debian trixie-updates InRelease
Hit:3 http://security.debian.org/debian-security trixie-security InRelease
All packages are up to date.
erwina@erwina:~$ sudo apt install openssh-server
openssh-server is already the newest version (1:10.0p1-7).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
erwina@erwina:~$
```

Screenshot successful execution SCP command:

```
erwina@erwina-VMware-Virtual-Platform: ~/Downloads
erwina@erwina-VMware-Virtual-Platform:/$ cd ~
erwina@erwina-VMware-Virtual-Platform:~$ cd ~/Downloads
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ scp test.txt erwina@192.168.139.131:/home/erwina/
The authenticity of host '192.168.139.131 (192.168.139.131)' can't be established.
ED25519 key fingerprint is SHA256:Cf20v7WvAkVCJQUL2uCwrZJgCWAN9ImbUDmI6RMgn6w.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? tes
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '192.168.139.131' (ED25519) to the list of known hosts.
erwina@192.168.139.131's password:
Permission denied, please try again.
erwina@192.168.139.131's password:
test.txt 100% 6 1.8KB/s 00:00
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ ls
apple2.jpg apple3.txt code decoded.gif email-base64.txt hello oldcar sherlock.txt simple test.txt
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ cat test.txt
hello
erwina@erwina-VMware-Virtual-Platform:~/Downloads$
```

Screenshot remmina:



Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:

```
Microsoft Windows [Version 10.0.26200.7462]  
(c) Microsoft Corporation. All rights reserved.
```

```
C:\Users\erwin>nslookup  
DNS request timed out.  
    timeout was 2 seconds.  
Default Server:  UnKnown  
Address:  2a02:a46c:2ed5:0:a7b:12ff:fea2:5570
```

```
> amazon.com  
Server:  UnKnown  
Address:  2a02:a46c:2ed5:0:a7b:12ff:fea2:5570
```

```
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
*** Request to UnKnown timed-out  
>
```

```
> google.com  
Server:  UnKnown  
Address:  2a02:a46c:2ed5:0:a7b:12ff:fea2:5570
```

```
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
DNS request timed out.  
    timeout was 2 seconds.  
*** Request to UnKnown timed-out  
>
```

```
> one.one.one.one
Server: UnKnown
Address: 2a02:a46c:2ed5:0:a7b:12ff:fea2:5570

DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
*** Request to UnKnown timed-out
> |
```

```
> dns.google.com
Server: UnKnown
Address: 2a02:a46c:2ed5:0:a7b:12ff:fea2:5570

DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
*** Request to UnKnown timed-out
>
```

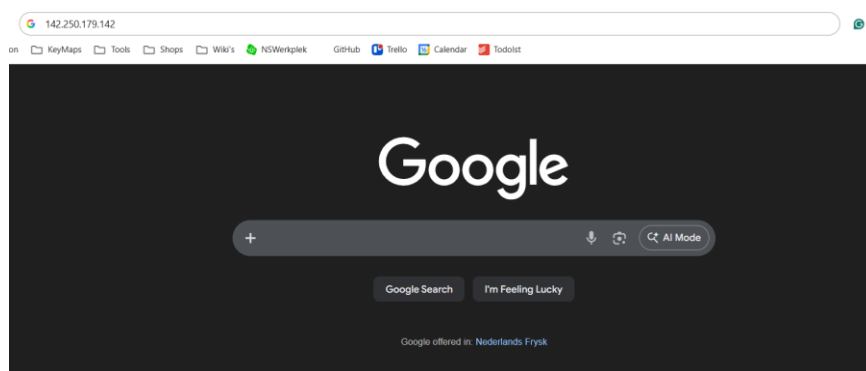
```
> w3schools.com
Server: UnKnown
Address: 2a02:a46c:2ed5:0:a7b:12ff:fea2:5570

DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
*** Request to UnKnown timed-out
> |
```

```
> bol.com
Server: UnKnown
Address: 2a02:a46c:2ed5:0:a7b:12ff:fea2:5570

DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
DNS request timed out.
  timeout was 2 seconds.
*** Request to UnKnown timed-out
> |
```

Screenshot website visit via IP address:



Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

- A /25 leaves 7 hosts bits (because $32 - 25 = 7$).
- So total addresses - $2^7 = 128$ addresses.

What is the usable IP range to hand out to the connected computers?

- A /25 means the subnet size is 128 addresses. So, the blocks in the last octet are:
 - 0-127
 - 128-255

The Network Address is 192.168.110.128, and the Broadcast address 192.168.110.255 is not useable. The useable range: 192.168.110.129 – 192.168.110.254, and useable host count: $128 - 2 = 126$

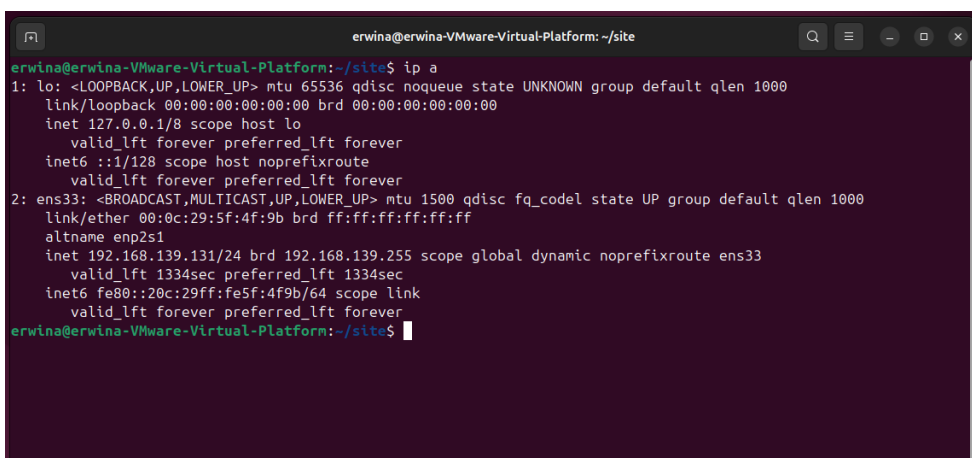
Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`

Explain the above calculation in your own words.

A /25 subnet contains 128 IP addresses, where the first is the network address and the last is the broadcast address, leaving 126 usable IPs for devices.

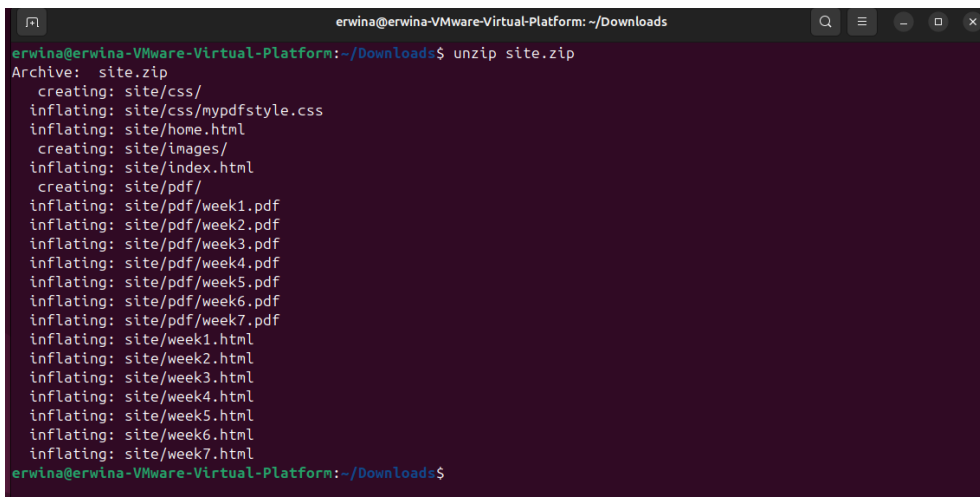
Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:

A terminal window titled 'erwina@erwina-VMware-Virtual-Platform: ~/site' showing the output of the 'ip a' command. The output displays details for the loopback interface 'lo' and the ethernet interface 'ens33'.

```
erwina@erwina-VMware-Virtual-Platform:~/site$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5f:4f:9b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.139.131/24 brd 192.168.139.255 scope global dynamic noprefixroute ens33
        valid_lft 1334sec preferred_lft 1334sec
    inet6 fe80::20c:29ff:fe5f:4f9b/64 scope link
        valid_lft forever preferred_lft forever
erwina@erwina-VMware-Virtual-Platform:~/site$
```

Screenshot of Site directory contents:

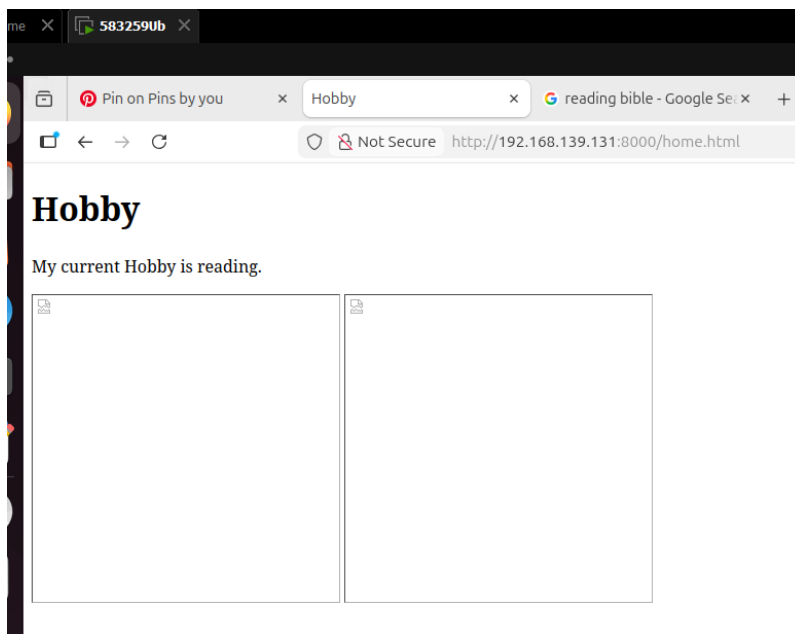
A terminal window titled 'erwina@erwina-VMware-Virtual-Platform: ~/Downloads' showing the output of the 'unzip site.zip' command. The output lists the files and directories being extracted from the archive.

```
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ unzip site.zip
Archive:  site.zip
  creating: site/css/
  inflating: site/css/mypdfstyle.css
  inflating: site/home.html
  creating: site/images/
  inflating: site/index.html
  creating: site/pdf/
  inflating: site/pdf/week1.pdf
  inflating: site/pdf/week2.pdf
  inflating: site/pdf/week3.pdf
  inflating: site/pdf/week4.pdf
  inflating: site/pdf/week5.pdf
  inflating: site/pdf/week6.pdf
  inflating: site/pdf/week7.pdf
  inflating: site/week1.html
  inflating: site/week2.html
  inflating: site/week3.html
  inflating: site/week4.html
  inflating: site/week5.html
  inflating: site/week6.html
  inflating: site/week7.html
erwina@erwina-VMware-Virtual-Platform:~/Downloads$
```

Screenshot python3 webserver command:

```
erwina@erwina-VMware-Virtual-Platform: ~/site
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ cd ~/site
bash: cd: ~/site: No such file or directory
erwina@erwina-VMware-Virtual-Platform:~/Downloads$ cd ~/site
erwina@erwina-VMware-Virtual-Platform:~/site$ nano home.html
erwina@erwina-VMware-Virtual-Platform:~/site$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default
    link/ether 00:0c:29:5f:4f:9b brd ff:ff:ff:ff:ff:ff
    inet 192.168.139.131/24 brd 192.168.139.255 scope global dynamic noprefixroute
        valid_lft 1658sec preferred_lft 1658sec
    inet6 fe80::20c:29ff:fe5f:4f9b/64 scope link
        valid_lft forever preferred_lft forever
erwina@erwina-VMware-Virtual-Platform:~/site$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
```

Screenshot web browser visits your site



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

Example: 192.168.1.100/27

Calculate the network segment

IP Address: 11000000.10101000.00000001.01100100

Subnet Mask: 11111111.11111111.11111111.11100000

Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.

For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).

The range of this network segment is from 192.168.1.96 to 192.168.1.127.

Paste source code here, with a screenshot of a working application.

```
Enter IP:
192.168.1.100
Enter Subnet Mask:
255.255.255.224

Calculate the network segment

IP Address: 11000000.10101000.00000001.01100100
Subnet Mask: 11111111.11111111.11111111.11100000
-----
Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.

For a /27 subnet, each segment (or subnet) has 32 IP addresses ( $2^5$ ).
The range of this network segment is from 192.168.1.96 to 192.168.1.127.
```

```
SaxionApp.println("Enter IP:");
String ip = SaxionApp.readString();

SaxionApp.println("Enter Subnet Mask:");
String mask = SaxionApp.readString();

String[] ipParts = ip.split("\\.");
String[] maskParts = mask.split("\\.");

String ipBinary = "";
String maskBinary = "";
String netBinary = "";

for (int i = 0; i < 4; i++) {
    int ipOct = Integer.parseInt(ipParts[i]);
    int maskOct = Integer.parseInt(maskParts[i]);
    int netOct = ipOct & maskOct;

    ipBinary += String.format("%8s", Integer.toBinaryString(ipOct)).replace(" ",
"0");
    maskBinary += String.format("%8s", Integer.toBinaryString(maskOct)).replace(" ",
"0");
    netBinary += String.format("%8s", Integer.toBinaryString(netOct)).replace(" ",
"0");

    if (i < 3) {
        ipBinary += ".";
    }
}
```

```

        maskBinary += ".";
        netBinary += ".";
    }
}

SaxionApp.println("");
SaxionApp.println("Calculate the network segment");
SaxionApp.println("");
SaxionApp.println("IP Address:  " + ipBinary);
SaxionApp.println("Subnet Mask:  " + maskBinary);
SaxionApp.println("-----");
SaxionApp.println("Network Addr: " + netBinary);
SaxionApp.println("");
SaxionApp.println("This gives 192.168.1.96 in decimal as the network address.");
SaxionApp.println("");
SaxionApp.println("For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).");
SaxionApp.println("The range of this network segment is from 192.168.1.96 to 192.168.1.127.");

```